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Spatial modelling of household and environmental correlates of inequitable WASH access in Ogun State, Nigeria

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Abstract

Background: Achieving universal and equitable access to water, sanitation, and hygiene (WASH) services is central to Sustainable Development Goal 6 (SDG 6). However, progress monitoring in sub-Saharan Africa remains constrained by the coarse spatial resolution of existing frameworks, such as WASHNORM, which provide only state-level estimates and are therefore insufficient for targeted local-level planning and intervention. To address this gap, we present the first high-resolution, local government area (LGA)-level spatial modelling of WASH conditions across Ogun State, integrating environmental and socioeconomic predictors within a machine learning framework.

Methods: We conducted a cross-sectional household survey involving 3,069 households across 88 communities spanning all 20 LGAs in Ogun State, Nigeria. Water, sanitation, and hygiene conditions were assessed using WHO/UNICEF Joint Monitoring Programme standards. A Random Forest classification model was developed using environmental predictors (rainfall and elevation) and socioeconomic predictors (population density, economic index, and degree of urbanization) to generate spatially continuous predictions of WASH conditions across the state.

Results: Although 68.4% of households had access to improved water sources, open defecation remained common (40.1%), and fewer than half (42.3%) had functional handwashing facilities, indicating substantial deficits in sanitation and hygiene relative to water access. The model demonstrated modest to moderate discriminatory performance across all WASH indicators and classes (AUC: 0.627–0.764). Spatial predictions identified 11 of the 20 LGAs as having critical or poor WASH conditions. Population density and economic index emerged as stronger predictors of WASH access than degree of urbanization, with middle- and upper-income areas consistently demonstrating better WASH outcomes across all components.

Conclusion: This study provides the first sub-district spatial baseline of WASH conditions across Ogun State and presents a replicable geospatial modelling framework for sub-national SDG 6 monitoring in Nigeria. The findings highlight the need

to prioritize densely populated and economically disadvantaged communities, particularly for sanitation and hygiene interventions, rather than relying solely on rural–urban classifications as the primary basis for WASH planning and resource allocation.

Keywords: WASH, Modelling, GIS, Water, Sanitation, Hygiene, West Africa

1 Background

Sustainable Development Goal 6 (SDG 6) aims to achieve universal and equitable access to safe drinking water, sanitation, and hygiene (WASH) services by 2030 [1]. However, progress toward this target remains uneven, particularly across low- and middle-income countries where inadequate WASH conditions continue to pose substantial public health risks [2]. Despite global improvements in water, sanitation and hygiene (WASH) coverage, substantial inequalities in access to safely managed services remain, particularly in sub-Saharan Africa. According to the WHO/UNICEF Joint Monitoring Programme (JMP) [3], billions of people worldwide still lack access to safely managed drinking water, sanitation and hygiene services, with the greatest burden occurring in low- and middle-income countries where rapid urbanization, population growth and infrastructural limitations continue to widen service gaps.

Poor access to safe water, sanitation, and hygiene is strongly associated with diarrheal diseases, respiratory infections, malnutrition, and several neglected tropical diseases, particularly among children under five years of age [4–7]. Beyond direct health effects, inadequate WASH also contributes to broader social and environmental inequalities, disproportionately affecting economically disadvantaged populations and underserved communities [8].

Access to safely managed WASH services remains far from universal. Improved water sources include protected wells, boreholes, piped water, rainwater, and packaged water, whereas improved sanitation facilities hygienically separate human excreta from contact through technologies such as flush toilets, ventilated improved pit latrines, and composting toilets [9, 10]. According to recent global estimates, approximately two billion people lacked safely managed drinking water services and 3.6 billion lacked safely managed sanitation services in 2021 [11]. In Nigeria, inadequate WASH infrastructure continues to contribute substantially to the burden of diarrheal and enteric diseases, particularly among children under five years [12, 13]. Recent cholera outbreaks reported across multiple Nigerian states further underscore persistent deficiencies in water and sanitation systems and highlight the urgent need for targeted WASH interventions [12, 14].

Effective investment and planning for WASH improvement require reliable, spatially explicit, and context-specific data [15, 16]. In Nigeria, the Water, Sanitation and Hygiene National Outcome Routine Mapping (WASHNORM) programme provides important State-level estimates for monitoring and resource mobilization [17]. However, these aggregated estimates are insufficient for identifying localized disparities and priority intervention areas within States. This limitation is particularly important in heterogeneous settings such as Ogun State, where rapid urbanization, socio-economic inequalities, and varying environmental conditions may strongly influence WASH access at community level.

Recent advances in geospatial and machine learning approaches have improved the ability to model and predict WASH conditions across large geographic areas. Previous studies have applied spatial modelling techniques to investigate inequalities in sanitation access, predict water service coverage, and identify high-risk underserved populations in sub-Saharan Africa and other low-resource settings [17–19]. However, most existing analyses remain limited to national or State-level summaries, with few studies providing high-resolution predictions capable of informing local decision-making. Existing studies also lack high-resolution spatial modelling at sub-LGA level in Ogun State.

To address this gap, this study assessed WASH conditions among 3,069 households across 88 communities representing all 20 Local Government Areas (LGAs) in Ogun State, Nigeria. We further integrated household survey data with environmental and socioeconomic predictors within a geospatial machine learning framework to generate high-resolution spatial predictions of WASH conditions across the State. Specifically, the study aimed to: (i) assess household-level access to improved water, sanitation, and hygiene services; (ii) identify key environmental and socioeconomic predictors associated with WASH access; and (iii) generate spatial risk maps to support targeted, evidence-based WASH interventions and sub-national SDG 6 monitoring in Ogun State.

2 Methods

2.1 Study area

Ogun State, located in southwestern Nigeria, comprises 20 Local Government Areas (LGAs), with Abeokuta serving as the State capital. The State has an estimated population of approximately 5 million people and is characterized by rapid urbanization and diverse ecological zones. Vegetation ranges from freshwater swamps and mangrove forests in the southeast to rainforest and woody Guinea savannah in the northwestern areas. Annual rainfall varies from approximately 900 mm in northern areas to 1600 mm along the coast. Major occupations include farming, timber logging, and trading [20].

2.2 Study design and sampling procedures

This cross-sectional community-based study was conducted between March 2022 and August 2024. A systematic grid-based sampling framework was adopted to ensure broad geographic representation across all 20 LGAs of Ogun State while minimizing spatial clustering bias. A 15 km grid was overlaid on the administrative map of Ogun State, and grid-cell centroids were generated. Communities nearest to each centroid were selected as sampling locations, resulting in the inclusion of 88 communities distributed across the State (S1 Dataset). This approach improved representation across diverse ecological, socioeconomic, and urban–rural settings.

2.3 Household selection and questionnaire administration

A total sampling approach was adopted within the selected communities, resulting in 3,069 households being surveyed across the 20 LGAs (S2 Dataset). Preliminary visits were conducted to obtain community approval, engage community leaders, and explain the objectives of the study prior to data collection. Within each selected community, all accessible and eligible households encountered during the survey period were approached for participation. Structured interviewer-administered questionnaires were

completed by household heads or available adult respondents aged 18 years and above who were willing to provide informed consent and adequately represent the household.

Households were excluded where respondents declined participation, were unavailable after repeated visits, or where questionnaire responses were substantially incomplete for WASH assessment variables. Non-response was minimized through repeat household visits and community-level sensitization prior to questionnaire administration. Only households with complete and valid WASH assessment records were included in the final analysis. Direct observations of WASH conditions were also conducted. Where discrepancies occurred between self-reported responses and field observations, priority was given to observations made by trained field personnel, although such discrepancies were infrequent.

The questionnaire consisted of two sections: demographic characteristics and household WASH conditions. Information collected included household water sources, sanitation facilities, latrine condition and availability, handwashing facilities, availability of soap, and waste disposal practices (S3 Questionnaire). To minimize interviewer bias, field assessments followed standardized WHO/UNICEF criteria for classifying improved and unimproved WASH conditions.

2.4 Scoring and classification of WASH conditions

The status of household WASH resources was assessed using criteria adapted from WHO/UNICEF standards and the framework described by Mogaji et al. [18]. Indicators meeting the criteria for improved WASH conditions were assigned a score of 1, whereas unimproved conditions were assigned a score of 0.

Each WASH component (water, sanitation, and hygiene) consisted of six equally weighted indicators, resulting in a maximum component score of six and an overall WASH score of 18. Equal weighting was adopted because all indicators represented essential components of household WASH access and no prior evidence justified differential weighting.

Households scoring ≥ 5 within a component category were classified as improved for that component, whereas scores < 5 were classified as unimproved. These thresholds were selected to represent substantial compliance with recommended WASH standards and to distinguish households with consistently adequate access and practices from those with partial or inadequate service conditions. Overall community-level WASH status was then determined based on the proportion of households classified as improved using the following categories: critical ($< 25\%$), poor (25–49.9%), fair (50–69.9%), and adequate (70–100%) (Table 1).

2.5 Environmental and socio-economic predictors

Environmental and socioeconomic predictor variables were obtained from publicly available geospatial datasets. Elevation and annual rainfall were obtained from the WorldClim database, whereas population density, degree of urbanization, and economic index datasets were obtained from the Socioeconomic Data and Applications Center (SEDAC) and the West Africa Coastal Vulnerability Mapping initiative [21–23]. These predictors were selected because of their established or potential influence on WASH access and infrastructure distribution [24].

Table 1 Scoring of WASH components at the community level

WASH Component	No of Items scored	Improved	Unimproved	Status
Overall WASH	18 (Combination of all components)	≥ 15	< 15	Critical = $< 25\%$ of surveyed household with improved conditions
Water	6 (Condition, Types, Source, Frequency, Location, Distance)	≥ 5	< 5	Poor = between 25 and 49% of the surveyed households with improved conditions
Sanitation	6 (Type, Condition, Gender segregated toilets, Presence of handwashing facilities, Open defecation and number of toilets)	≥ 5	< 5	Fair = between 50 and 69% of the surveyed households with improved conditions Adequate = between 70 and 100% of the surveyed households with improved conditions
Hygiene	6 (Washing of hand, presence of refuse dump sites, presence of domestic animals, regular use of footwear, trimming of fingernails, material used for anal cleaning)	≥ 5	< 5	

All raster predictors were resampled to a common spatial resolution of 1 km $\times$ 1 km. This resolution was selected to balance spatial detail with computational efficiency and to align with the resolution of several environmental covariates used in the modelling framework. The selected resolution also enabled high-resolution subdistrict-level prediction while minimizing excessive spatial noise and computational burden.

2.6 Spatial modelling

To predict the spatial distribution of WASH conditions across Ogun State, we employed a Random Forest (RF) classification model. Random Forest was selected because it performs well with high-dimensional environmental datasets, is robust to multicollinearity, and effectively captures complex nonlinear relationships and interactions between predictors and response variables [25]. In addition, RF models are less sensitive to overfitting compared with single-tree approaches because predictions are derived from aggregated ensemble outputs.

Predictor variables were standardized before model fitting to ensure comparability across variables. The response variable represented four ordered WASH adequacy classes ranging from critical (1) to adequate (4). The RF model consisted of 500 decision trees. The use of 500 trees was chosen to ensure model stability and reduce prediction variance while maintaining computational efficiency.

2.7 Model performance evaluation

Model performance was evaluated using five-fold spatial block cross-validation implemented in the blockCV package in R [26]. Block sizes were defined using the spatial autocorrelation range of the raster covariates to ensure spatial separation between training and testing folds. Model discriminatory performance was assessed using one-versus-rest Area Under the Receiver Operating Characteristic Curve (AUC) values computed for each WASH adequacy class across folds, with mean AUC and standard deviation

used to summarize model accuracy and stability. To further reduce sampling variability, the modelling procedure was repeated across 10 replicates and predictions were averaged across runs. AUC values range from 0.5 (random classification) to 1.0 (perfect discrimination), with higher values indicating stronger predictive performance.

SHAP (SHapley Additive exPlanations) bee-swarm summary plots were generated to evaluate the relative contribution and directional influence of environmental and socio-demographic predictors on the probability of belonging to the adequate WASH conditions category across the Overall WASH, WATER, SANITATION and HYGIENE Random Forest models. In addition, the modelling framework generated pixel-level uncertainty estimates based on entropy derived from Random Forest class probabilities, highlighting areas where prediction confidence varied due to local heterogeneity in environmental conditions and sampling density.

2.8 Methodological considerations and limitations

Several limitations should be considered when interpreting these findings. First, although the grid-based sampling framework improved geographic representativeness, some degree of sampling bias may remain because community accessibility and participation could influence household inclusion. Second, parts of the questionnaire relied on self-reported household information, which may introduce reporting bias. Third, while the 1 km spatial resolution enabled high-resolution prediction at subdistrict scale, finer-scale heterogeneity in WASH conditions may not have been fully captured. Finally, because the study employed a cross-sectional design, observed associations should not be interpreted as causal relationships.

3 Results

3.1 Summary of household participant demographics

A total of 3069 households were surveyed across the 88 selected communities in Ogun State. Most respondents were between 31 and 42 years of age, and slightly more males than females participated in the survey. Household sizes were generally moderate, with more than half of households comprising 4–6 members (S4 Table). These demographic characteristics suggest that the study captured a broad representation of economically active households across the State.

3.2 Household water, sanitation, and hygiene conditions

Access to improved water sources was relatively high, with 68.4% (95% CI: 66.7–70.0%) of households reporting improved water access, although hand-dug wells remained the dominant source of water supply across the State (Table 2). Despite this relatively high coverage, more than half of households lacked on-site water access, indicating continued reliance on off-premise water collection. This finding suggests that physical access to water remains a challenge despite the apparent availability of improved sources.

Sanitation conditions were substantially poorer than water access indicators. More than 40% of households lacked improved sanitation facilities and practiced open defecation, while gender-segregated toilets and soap availability within toilets were uncommon (Table 3). In addition, most households with toilets had fewer than two facilities per household, and many toilets were classified as poorly maintained. Together, these

Table 2 status and condition of water resources in the 88 surveyed communities

Variables	Frequency (N = 3069)	Percentage	95% CI
Condition of water source			
Unimproved	970	31.6	30.0–33.3
Improved	2099	68.4	66.7–70.0
Type of water source			
Well	2791	90.9	89.9–91.9
Bore-hole	160	5.2	4.4–6.0
Pipe-borne Water	73	2.4	1.9–3.0
Others	45	1.5	1.1–1.9
Location of water source			
Offsite	1657	54	52.2–55.7
Onsite	1412	46	44.3–47.8
Frequency of supply			
Non-daily	831	27.1	25.5–28.7
Daily	2238	72.9	71.3–74.5
Condition of water supply			
Abandoned	68	2.2	1.7–2.8
Functioning	3001	97.8	97.2–98.3
Distance to water source			
Outside 5 min	868	28.3	26.7–29.9
Within 5 min	2201	71.7	70.1–73.3
Complaint when sourcing water			
Yes	789	25.7	24.2–27.3
No	2280	74.3	72.7–75.8

Table 3 Status and condition of sanitation resources in the 88 surveyed communities

Variables	Frequency (N = 3069)	Percentage	95% CI
Type of sanitation facility			
Unimproved	1326	43.2	41.4–44.9
Improved	1743	56.8	55.1–58.6
Open Defecation due to lack of toilet facility			
Yes	1231	40.1	38.4–41.8
No	1838	59.9	58.2–61.6
Number of toilets (n = 1838 households with toilets)			
Less than 2 per household	1496	81.4	79.5–83.1
At least 2 per household	342	18.6	16.9–20.5
Presence of gender segregated toilet (n = 1838)			
No	1529	83.2	81.4–84.9
Yes	309	16.8	15.1–18.6
Condition of toilets (n = 1838)			
Dirty	1097	59.7	57.4–61.9
Clean	741	40.3	38.1–42.6
Usage of toilet (n = 1838)			
Abandoned	88	4.8	3.9–5.9
Functioning	1750	95.2	94.1–96.1
Presence of soap in the toilet (n = 1838)			
No	1763	95.9	94.9–96.7
Yes	75	4.1	3.3–5.1

findings indicate that sanitation infrastructure and hygiene support facilities remain major gaps in household WASH conditions across Ogun State.

Hygiene indicators showed mixed patterns (Table 4). Although most respondents reported washing their hands after toilet use, fewer than half of households had functional handwashing facilities within toilets. Refuse dumpsites were also common within household premises, and several risky hygiene behaviors, including irregular footwear use and fingernail biting, were frequently reported. These patterns suggest a discrepancy between self-reported hygiene practices and the availability of enabling hygiene infrastructure.

3.3 Observed spatial distribution of WASH conditions

The observed spatial distribution of WASH conditions revealed substantial heterogeneity across Ogun State (Fig. 1). Overall WASH conditions in most communities ranged from poor to fair, with only a limited number of communities classified

Table 4 Status and conditions of household hygiene resources in the 88 surveyed communities

Variables	Frequency (N = 3069)	Percentage	95% CI
Presence of hand-washing facilities in toilet			
No	1771	57.7	55.9–59.4
Yes	1298	42.3	40.6–44.1
Washing of hands after toilet use			
No	494	16.1	14.8–17.4
Yes	2575	83.9	82.6–85.2
Presence of domestic animals			
Yes	1562	50.9	49.1–52.6
No	1507	49.1	47.4–50.9
Presence of refuse dumpsite within the premises			
Yes	2139	69.7	68.1–71.3
No	930	30.3	28.7–31.9
Sucking of fingers			
Yes	25	0.8	0.6–1.2
No	3044	99.2	98.8–99.4
Biting of fingernails			
Yes	1000	32.6	31.0–34.3
No	2069	67.4	65.7–69.0
Picking of food from the ground			
Yes	482	15.7	14.5–17.1
No	2587	84.3	82.9–85.5
Trimming of fingernails regularly			
Yes	1897	61.8	60.1–63.5
No	1172	38.2	36.5–39.9
Regular usage of footwear			
No	1277	41.6	39.9–43.4
Yes	1792	58.4	56.6–60.1
Materials used for anal cleaning			
Stick, leaf, paper or stone	1037	33.8	32.1–35.5
Water	2032	66.2	64.5–67.9

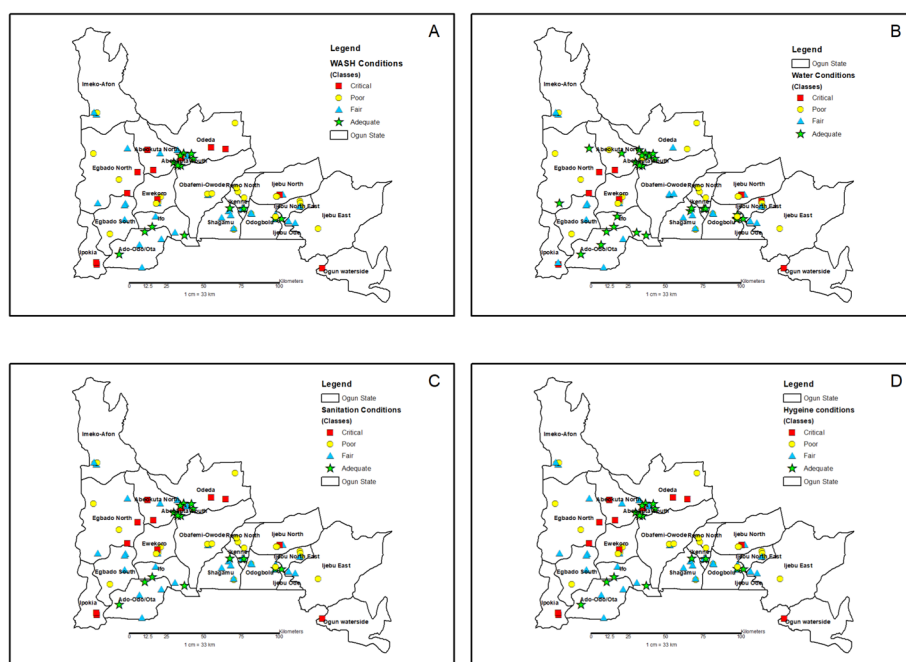


Fig. 1 Overall water, sanitation and hygiene conditions of the surveyed communities. **a** Overall WASH. **b** Water. **c** Sanitation. **d** Hygiene

as adequate. Adequate WASH conditions were concentrated primarily within more urbanized LGAs, including Abeokuta South, Ifo, Sagamu, and Ado-Odo/Ota, whereas several northern and rural LGAs showed persistent deficiencies.

Water access generally showed better conditions than sanitation and hygiene. In contrast, sanitation and hygiene indicators remained predominantly within the poor-to-fair categories across many communities. This pattern highlights a persistent imbalance in WASH development, where improvements in water access have not been matched by comparable progress in sanitation and hygiene infrastructure.

3.4 Predicted WASH conditions across Ogun State

Spatial prediction modelling identified marked geographic inequalities in WASH access across Ogun State (Fig. 2). Large areas of Odeda, Abeokuta North, Obafemi-Owode, Ipokia, Ewekoro, and Imeko-Afon were predicted to have critical overall WASH conditions, indicating that fewer than 25% of households in these areas had improved WASH access.

Predicted water conditions were generally better in southern LGAs, whereas sanitation and hygiene conditions remained poor across substantial portions of the State. Abeokuta South consistently demonstrated adequate predicted conditions across all WASH components. The spatial clustering of poor WASH outcomes in economically disadvantaged and less densely developed areas suggests that WASH inequalities are strongly localized rather than uniformly distributed across the State.

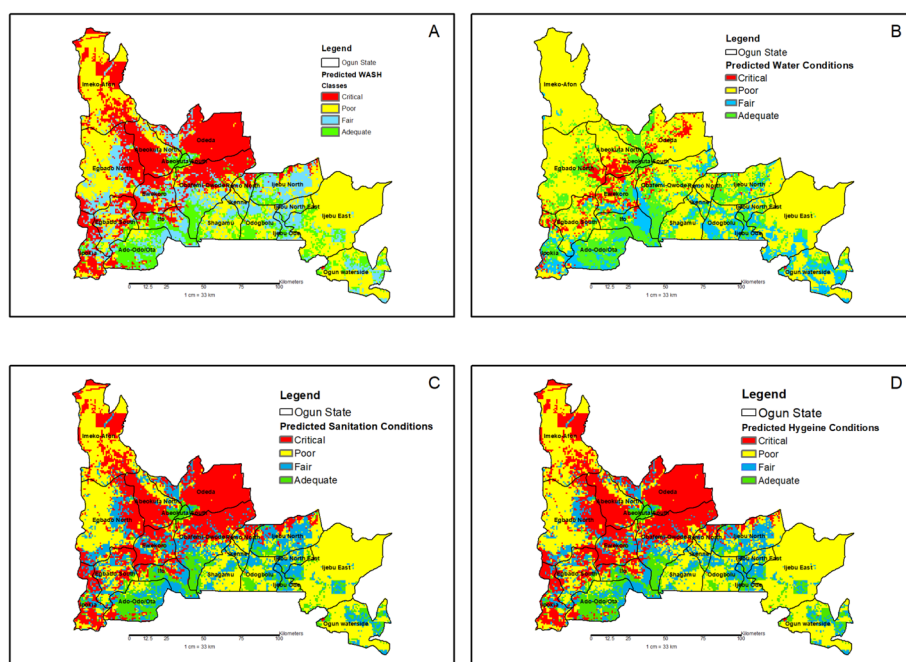


Fig. 2 Predicted water sanitation and hygiene outcome for Ogun State using Random Forest modeling approach. **a** Overall WASH. **b** Water. **c** Sanitation. **d** Hygiene

3.5 Model performance and uncertainty

The Random Forest model demonstrated modest to moderate predictive performance across most WASH classes, with AUC values generally ranging from 0.627 to 0.764 (Table 5). Although predictive performance was slightly lower for some “fair” and “poor” classes, all models performed better than random classification. These findings indicate that the selected environmental and socioeconomic predictors were effective in capturing major spatial patterns in WASH conditions across Ogun State.

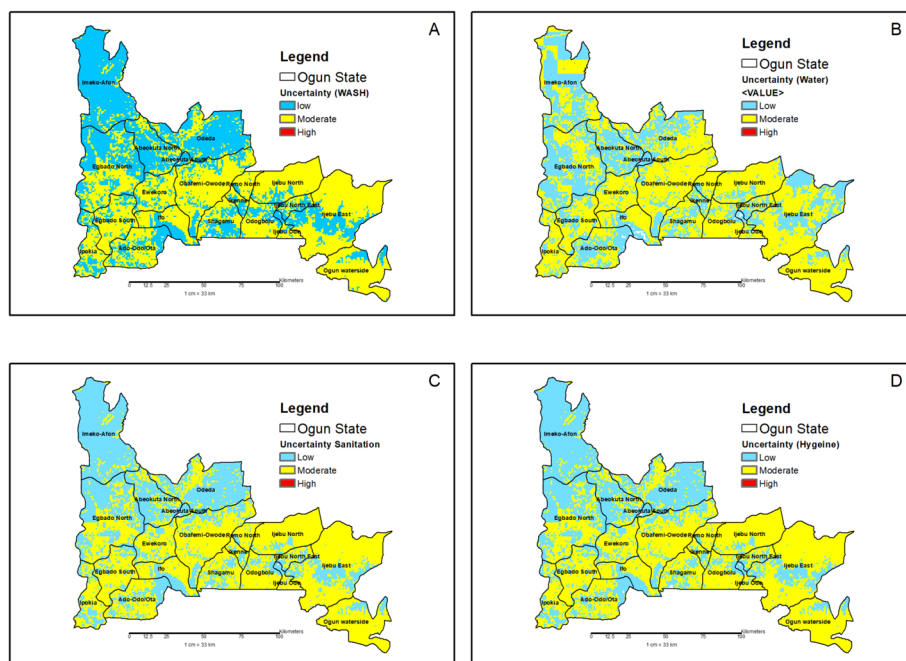
Uncertainty analysis showed that most areas exhibited low-to-moderate prediction uncertainty (Fig. 3). Higher uncertainty was mainly concentrated in localized areas (occurring more in the eastern parts of the State) with greater environmental heterogeneity or lower sampling density, indicating areas where additional field sampling may further improve prediction accuracy.

3.6 Predictor importance and environmental relationships

Predictor importance varied across WASH components (Table 6). Socio-economic Index was consistently important across all models, while rainfall contributed strongly to the Overall WASH, SANITATION, and HYGIENE models. In contrast, Degree of Urbanization and Population Density were particularly influential in predicting WATER access. SHAP analysis broadly supported these patterns (Fig. 4), showing that higher Socio-economic Index and Degree of Urbanization were generally associated with an increased probability of adequate WASH conditions. Collectively, these findings indicate that WASH inequalities are shaped by multiple socio-economic, settlement, and environmental factors rather than by rural–urban classification alone.

Table 5 Area under the curve (AUC) values for each of the Random Forest predictions for WASH, water, sanitation and Hygiene conditions in Ogun State using spatial cross validation method

Model	Class	Area Under the Curve (AUC)			
		Mean	SD	Minimum	Maximum
Overall WASH	Critical	0.723	0.205	0.536	1
	Poor	0.655	0.068	0.546	0.713
	Fair	0.672	0.148	0.52	0.917
	Adequate	0.747	0.085	0.64	0.849
WATER	Critical	0.666	0.134	0.526	0.794
	Poor	0.669	0.139	0.493	0.875
	Fair	0.627	0.058	0.561	0.702
	Adequate	0.697	0.121	0.514	0.8
SANITATION	Critical	0.69	0.239	0.464	1
	Poor	0.647	0.125	0.538	0.855
	Fair	0.635	0.184	0.407	0.917
	Adequate	0.764	0.085	0.64	0.829
HYGIENE	Critical	0.69	0.239	0.464	1
	Poor	0.647	0.125	0.538	0.855
	Fair	0.635	0.184	0.407	0.917
	Adequate	0.764	0.085	0.64	0.829

**Fig. 3** Level of uncertainties for all predicted WASH components. **a** Overall WASH. **b** Water. **c** Sanitation. **d** Hygiene

4 Discussion

This study examined the spatial distribution and determinants of household water, sanitation, and hygiene (WASH) conditions across Ogun State, Nigeria, and

Table 6 Relative permutation importance (%) of environmental and socio-demographic predictors across the Overall WASH, WATER, SANITATION and HYGIENE Random Forest models

Variable	Permutation importance (%)			
	Overall WASH	HYGIENE	SANITATION	WATER
Precipitation (mm)	37.3	37.3	37.3	5.6
Elevation (m)	1.8	1.8	1.8	0.8
Socio-economic Index	46.9	46.9	46.9	36.0
Population Density	8.7	8.7	8.7	27.2
Degree of Urbanization	5.2	5.2	5.2	30.3

Permutation importance values were standardized to relative percentages within each model based on the absolute contribution of each predictor to model performance

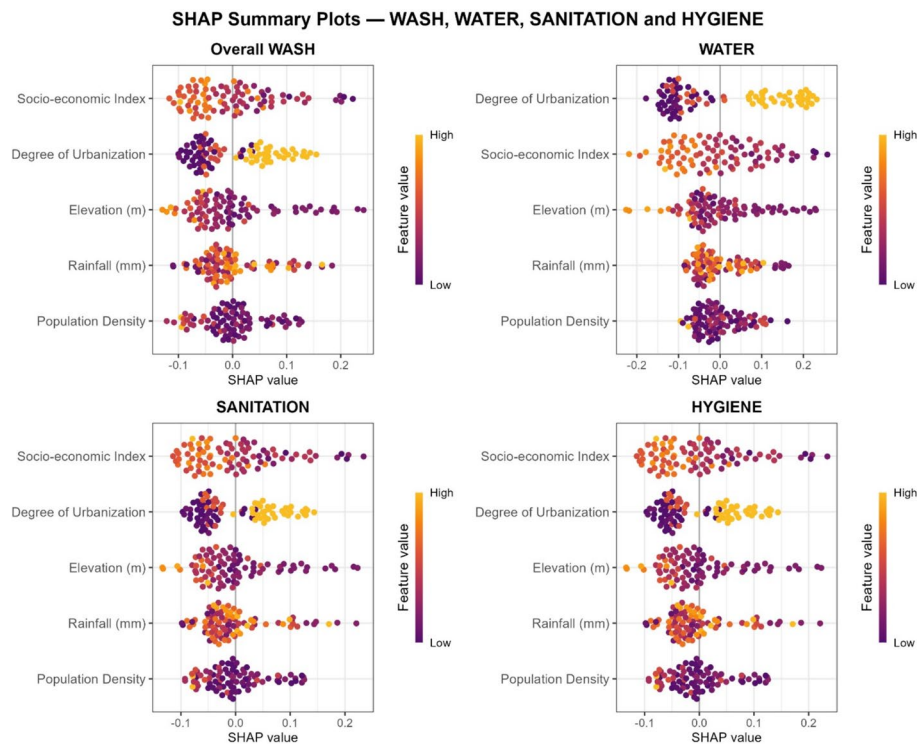


Fig. 4 SHAP bee-swarm summary plots illustrating the contribution of environmental and socio-demographic covariates to adequate WASH conditions across the Overall WASH, WATER, SANITATION and HYGIENE models

quantified localized inequalities using a geospatial machine learning framework.

The findings of the present study should be interpreted within the context of the 2021 Water Sanitation and Hygiene National Outcome Routine Mapping [17] survey, Nigeria’s national benchmarking framework for SDG 6 monitoring. Nationally, WASHNORM 2021 reported that 23% of Nigerians lacked access to basic water services, 48 million people still practiced open defecation, and only 10% of the population had access to combined basic water, sanitation and hygiene services [17]. Although Ogun State, located within the Southwest geopolitical zone, generally performed better than the national average for improved water access, the present study

identified persistent sanitation and hygiene deficiencies across several LGAs, including widespread open defecation and inadequate handwashing facilities.

These findings are consistent with WASHNORM 2021 reports showing that, despite relatively high-water access in the Southwest, sanitation and hygiene indicators remain comparatively poor within the region [17]. Furthermore, while WASHNORM provides important state-level estimates, the present geospatial modelling framework extends these findings by identifying fine-scale spatial heterogeneity and localized WASH inequalities across Ogun State that may remain obscured within aggregated state-level summaries.

The observed imbalance between water access and sanitation-hygiene conditions is consistent with previous studies from Nigeria and other low-resource settings, where investments in water infrastructure have frequently outpaced improvements in sanitation and hygiene services [27, 28]. Despite the relatively high proportion of households reporting improved water sources, reliance on off-premise water access and hand-dug wells remained common, suggesting that availability of improved sources does not necessarily translate into equitable or reliable household access.

Sanitation indicators revealed persistent deficiencies across much of the State, including widespread open defecation, inadequate toilet infrastructure, and limited availability of soap and handwashing facilities. These patterns are particularly important because sanitation and hygiene conditions are closely linked to the transmission dynamics of diarrheal diseases, enteric infections, and several environmentally mediated neglected tropical diseases [19, 29–37]. The persistence of these deficits despite moderate water access further suggests that sanitation and hygiene improvements may face distinct infrastructural, behavioural, and socioeconomic constraints [38].

One of the principal findings of this analysis was the strong spatial heterogeneity in WASH conditions across Ogun State. Spatial predictions identified clusters of persistently poor or critical WASH conditions concentrated within several less economically advantaged LGAs. Similar localized inequalities have been documented in previous geospatial WASH analyses conducted in other Nigeria, where broad administrative averages frequently mask substantial intra-regional disparities [19, 24]. Our findings therefore reinforce the importance of high-resolution spatial monitoring approaches capable of identifying underserved populations that may otherwise remain overlooked within conventional State-level reporting frameworks.

The modelling framework further identified population density and economic index as stronger predictors of WASH conditions than degree of urbanization. In this study, population density represented the concentration of people within a given area, whereas the urbanization predictor reflected the extent of built-up settlement and urban land-use characteristics derived from geospatial settlement datasets. This finding suggests that socioeconomic disadvantage and infrastructure pressure associated with rapid population growth may exert greater influence on WASH inequalities than simple rural–urban classifications alone. Although urbanization is frequently used as a proxy for WASH development, the present findings indicate that substantial heterogeneity persists within both urban and rural settings. Similar relationships between poverty, infrastructure deficits, and inadequate WASH access have been reported in previous spatial analyses from other low-resource settings [24].

This study also contributes methodologically by integrating household survey data with Random Forest-based spatial prediction to generate high-resolution subdistrict WASH estimates across Ogun State. Existing monitoring frameworks such as WASH-NORM provide valuable State-level indicators but remain insufficient for identifying fine-scale disparities required for targeted intervention planning. By generating continuous spatial predictions, the present framework provides a scalable approach for supporting sub-national SDG 6 monitoring and evidence-based allocation of WASH resources. The modelling framework used in this study is potentially scalable to other data-scarce settings at sub-national or national levels, provided that representative household survey data and appropriately resolved environmental and socioeconomic covariates are available. In geographically heterogeneous settings, finer-scale modelling may better capture localized WASH inequalities, whereas broader-scale analyses may be more suitable where environmental and socioeconomic conditions are relatively uniform.

5 Study limitations

This analysis should be interpreted within the context of its design and assumptions. First, although the grid-based sampling framework improved geographic coverage across Ogun State, some degree of sampling bias may remain because accessibility and household participation varied across communities. Second, several household indicators relied on self-reported responses and may therefore be subject to reporting bias. Third, while the 1 km spatial resolution enabled relatively detailed prediction, finer-scale heterogeneity in WASH conditions may not have been fully captured. Finally, because the study employed a cross-sectional design, the observed relationships between WASH conditions and environmental or socioeconomic predictors should be interpreted as associations rather than causal effects.

Another limitation of this study is that household data collection was conducted over an extended period from March 2022 to August 2024, during which seasonal variations in rainfall patterns, water availability, and potential changes in WASH infrastructure may have occurred across communities. Although the modelling framework incorporated environmental covariates, the analysis did not explicitly account for temporal or seasonal dynamics during the survey period. Consequently, some observed spatial patterns may partially reflect temporal variability in WASH access and environmental conditions across the study area. Future studies incorporating longitudinal or season-specific assessments may provide improved understanding of temporal changes in household WASH conditions.

Furthermore, WASH classification was based on observed household infrastructure and reported practices and did not include microbiological or physicochemical assessment of water quality. In addition, although the survey achieved broad geographic coverage across Ogun State, the number of training locations available for spatial modelling may have limited the ability of the models to fully capture local-scale environmental heterogeneity.

Taken together, these findings demonstrate that WASH inequalities across Ogun State remain strongly structured by localized socioeconomic and environmental conditions, particularly for sanitation and hygiene services. The results further emphasize the need to move beyond coarse administrative summaries toward high-resolution spatial

surveillance frameworks capable of supporting targeted, climate-resilient, and equity-focused WASH interventions.

6 Conclusion

This study provides the first high-resolution spatial assessment of household water, sanitation, and hygiene (WASH) conditions across Ogun State, Nigeria, integrating household survey data with geospatial machine learning approaches to identify localized inequalities in WASH access. Although access to improved water sources was relatively widespread, substantial sanitation and hygiene deficits persisted across many LGAs, with strong spatial heterogeneity observed throughout the State.

The findings demonstrate that WASH inequalities are more strongly associated with localized socioeconomic and population-related factors than with conventional rural–urban classifications alone. Spatial prediction modelling further identified several underserved LGAs with persistently critical or poor WASH conditions, emphasizing the limitations of broad State-level summaries for intervention planning.

Methodologically, this study demonstrates the utility of combining household-level WASH surveys with Random Forest-based spatial prediction to generate fine-scale, policy-relevant WASH estimates in data-limited settings. This framework provides a scalable approach for supporting sub-national SDG 6 monitoring and identifying priority communities for targeted intervention.

Improving WASH outcomes in Ogun State will likely require more geographically targeted and equity-focused strategies, particularly for sanitation and hygiene infrastructure in densely populated and economically disadvantaged communities. Future efforts should prioritize routine high-resolution spatial monitoring, integration of geospatial tools into WASH planning frameworks, and strengthened investment in underserved areas to support sustainable and climate-resilient WASH systems.

Abbreviation

WASH	Water Sanitation and Hygiene
LGA	Local Government Area
WHO	World Health Organization
UNICEF	United Nations Children's Fund
GIS	Geographic Information System
SDG	Sustainable Development Goal
WASHNORM	Water Sanitation and Hygiene National Outcome Routine Mapping
GE	Google Earth
GPS	Global Positioning System
SRTM	Shuttle Radar Topography Mission
SEDAC	Socioeconomic Data Applications Centre
SMOD	Settlement Model Grid Data
GDAL	Geospatial Data Abstraction Library
RF	Random Forest
AUC	Area Under the Receiver Operating Characteristic Curve
ROC	Receiver Operating Characteristic Curve

Supplementary Information

The online version contains supplementary material available at <https://doi.org/10.1007/s44274-026-00857-3>.

Supplementary file 1.
Supplementary file 2.
Supplementary file 3.
Supplementary file 4.

Supplementary file 5.

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Author contribution

Conceptualization—AS. Supervision—AS, MA, HO, OA, LO. Investigation—AS, OA, TA, OA, OA, A Software and visualization—AS. Data collection—AS, LO, B, AO, AO, JK, OF, TA, OA, F, AB, CJ, O, AO, VP, D, K, AM, TP, OA, AO, MA, OA. Formal analysis—AS, AB, OA. Data curation—AS, OA. Writing original draft—AS. Writing review and editing—AS, LO, B, HO, AO, JK, OF, TA, OA, F, AB, CJ, O, AO, VP, D, K, AM, TP, OA, AO, MA, OA

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Data availability

All data generated and analyzed during the study are present in the article.

Declarations

Ethics approval and consent to participate

The ethical review committee of the Fountain University Center for Research, Innovation and Technology, Osogbo, Osun State, Nigeria granted approval to carry out this work (Reference No: FUO/ERB/24/25/0036). The study adhered to the Declaration of Helsinki. Informed consent was obtained from all participants prior to data collection. Participants were informed about the study objectives, procedures, potential risks and benefits, and their rights as participants. Participation was voluntary, and only individuals who agreed to participate and provided informed consent were enrolled in the study.

Consent for publication

Not applicable.

Competing interests

The authors declare no competing interests.

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